Chapter 1. Vector Spaces

Linear combinations, independence, bases and dimension

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1.1 Introduction

Motivation. Forces and velocities add by the parallelogram law and can be stretched by real factors. Eight algebraic rules satisfied by these two operations are all that linear algebra uses, so any system obeying them (tuples, matrices, functions, polynomials) can be treated by the same theory.

Two vectors x, y acting at a point P add by the **parallelogram law**: $x + y$ is the diagonal of the parallelogram with sides x and y. With P at the origin, if x ends at (a_1, a_2) and y at (b_1, b_2), then $x + y$ ends at $(a_1 + b_1, a_2 + b_2)$, and tx ends at (ta_1, ta_2).

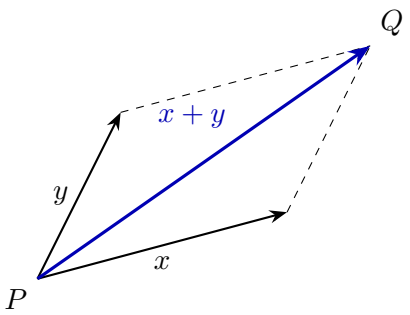


Figure. *Parallelogram law: $x + y$ is the diagonal through P.*

Geometric vectors satisfy, for all vectors x, y, z and reals a, b:

1. $x + y = y + x$; 2. $(x + y) + z = x + (y + z)$; 3. there is 0 with $x + 0 = x$; 4. each x has y with $x + y = 0$;

5. $1x = x$; 6. $(ab)x = a(bx)$; 7. $a(x + y) = ax + ay$; 8. $(a + b)x = ax + bx$.

Lines and planes. If u, v end at A, B, the line through A and B is $x = u + t(v - u)$, $t \in \mathbb{R}$. If u, v start at A and end at B, C (three noncollinear points), the plane through A, B, C is $x = A + su + tv$, $s, t \in \mathbb{R}$.

Example 1

For $A = (-2, 0, 1)$, $B = (4, 5, 3)$: $B - A = (6, 5, 2)$, and the line is $x = (-2, 0, 1) + t(6, 5, 2)$.

Example 2

For $A = (1, 0, 2)$, $B = (-3, -2, 4)$, $C = (1, 8, -5)$: $B - A = (-4, -2, 2)$, $C - A = (0, 8, -7)$, and the plane is $x = (1, 0, 2) + s(-4, -2, 2) + t(0, 8, -7)$.

1.2 Vector spaces

Motivation. The definition abstracts the eight rules. Many familiar collections (tuples, matrices, functions, polynomials, sequences) satisfy them, and a handful of consequences, such as cancellation and uniqueness of 0 and $-x$, follow from the axioms alone.

Definition (Vector space)

A **vector space** V over a field $\mathbb{F}$ (see Appendix C) is a set with an addition $V \times V \to V$ and a scalar multiplication $\mathbb{F} \times V \to V$ such that for all $x, y, z \in V$, $a, b \in \mathbb{F}$:
(VS 1) $x + y = y + x$; (VS 2) $(x + y) + z = x + (y + z)$;
(VS 3) there is $0 \in V$ with $x + 0 = x$ for all x; (VS 4) for each x there is y with $x + y = 0$;
(VS 5) $1x = x$; (VS 6) $(ab)x = a(bx)$;
(VS 7) $a(x + y) = ax + ay$; (VS 8) $(a + b)x = ax + bx$.
Elements of $\mathbb{F}$ are **scalars**, elements of V **vectors**.

An n**-tuple** $(a_1, \ldots, a_n)$ has entries in $\mathbb{F}$; tuples are equal iff all entries agree. An $m \times n$ **matrix** A has entries A_{ij} (i-th row, j-th column); A_{ii} are the **diagonal entries**, O is the **zero matrix**, and A is **square** if $m = n$.

Example 1

$\mathbb{F}^n$, the n-tuples with coordinatewise operations, is a vector space; e.g. in $\mathbb{C}^2$, $(1+i, 2)+(2-3i, 4i) = (3-2i, 2+4i)$ and $i(1+i, 2) = (-1+i, 2i)$. Vectors of $\mathbb{F}^n$ are often written as columns; $\mathbb{F}^1 = \mathbb{F}$.

Example 2

$\mathrm{M}_{m\times n}(\mathbb{F})$, with $(A + B)_{ij} = A_{ij} + B_{ij}$ and $(cA)_{ij} = cA_{ij}$, is a vector space.

Example 3

For a nonempty set S, the functions $\mathcal{F}(S, \mathbb{F})$ with $(f+g)(s) = f(s)+g(s)$ and $(cf)(s) = cf(s)$ form a vector space.

A **polynomial** $f(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_k \in \mathbb{F}$ has **degree** the largest k with $a_k \neq 0$; the **zero polynomial** has degree -1. Over an infinite field we may regard polynomials as functions $\mathbb{F} \to \mathbb{F}$.

Example 4

$\mathsf{P}(\mathbb{F})$, the polynomials with coefficientwise addition and scalar multiplication (pad the shorter one with zero coefficients), is a vector space.

Example 5

The sequences $\{a_n\}$ in $\mathbb{F}$ with only finitely many nonzero terms, with termwise operations, form a vector space.

Example 6

On $S = \mathbb{R}^2$ with $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 - b_2)$ and $c(a_1, a_2) = (ca_1, ca_2)$, (VS 1), (VS 2) and (VS 8) fail: S is not a vector space.

Example 7

On $S = \mathbb{R}^2$ with $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, 0)$ and $c(a_1, a_2) = (ca_1, 0)$, (VS 3) (hence (VS 4)) and (VS 5) fail.

Theorem 1.1 (Cancellation law for vector addition)

If $x + z = y + z$ in V, then $x = y$.

Proof. Take v with $z + v = 0$ (VS 4). Then $x = x + 0 = x + (z + v) = (x + z) + v = (y + z) + v = y + (z + v) = y + 0 = y$ by (VS 2), (VS 3). ■

Corollary 1

The vector 0 of (VS 3) is unique.

Proof [Supplement]. If $0'$ also satisfies (VS 3), then $0 + 0' = 0 = 0 + 0$ (by (VS 1), (VS 3)), so $0' = 0$ by Theorem 1.1. ■

Corollary 2

The vector y of (VS 4) is unique.

Proof [Supplement]. If $x+y = 0 = x+y'$, then $y+x = y'+x$ by (VS 1), so $y = y'$ by Theorem 1.1. ■

0 is the **zero vector**, and the y of (VS 4) is the **additive inverse** $-x$.

Theorem 1.2

In any vector space V:
(a) $0x = 0$ for each $x \in V$.
(b) $(-a)x = -(ax) = a(-x)$ for each $a \in \mathbb{F}$, $x \in V$.
(c) $a0 = 0$ for each $a \in \mathbb{F}$.

Proof.

(a) $0x + 0x = (0 + 0)x = 0x = 0x + 0 = 0 + 0x$, so $0x = 0$ by Theorem 1.1.

(b) $ax+(-a)x = (a+(-a))x = 0x = 0$, so $(-a)x = -(ax)$ by Corollary 2. In particular $(-1)x = -x$, and $a(-x) = a((-1)x) = (a(-1))x = (-a)x$ by (VS 6).

(c) [Supplement] $a0 + a0 = a(0 + 0) = a0 = a0 + 0$, so $a0 = 0$ by Theorem 1.1. ■

1.3 Subspaces

Motivation. A subset that is itself a vector space under the inherited operations is a subspace. Three closure checks suffice. Intersections of subspaces are subspaces; unions usually are not, which is why sums are introduced.

Definition (Subspace)

A subset W of V is a **subspace** if it is a vector space over $\mathbb{F}$ with the operations of V. V and the **zero subspace** $\{0\}$ are subspaces.

(VS 1, 2, 5–8) are inherited by any subset, so W is a subspace iff it is closed under addition and scalar multiplication, has a zero vector, and contains additive inverses.

Theorem 1.3

$W \subseteq V$ is a subspace iff (a) $0 \in W$,
(b) $x + y \in W$ for $x, y \in W$, and (c) $cx \in W$ for $c \in \mathbb{F}$, $x \in W$.

Proof.

Step 1 (only if). (b), (c) hold since W is a vector space under these operations. If $0'$ is the zero of W, then $x + 0' = x = x + 0$ for $x \in W$, so $0' = 0$ by Theorem 1.1, and $0 \in W$.

Step 2 (if). It remains to see that inverses lie in W: for $x \in W$, $-x = (-1)x \in W$ by (c) and Theorem 1.2. ■

The **transpose** A^t of $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ has $(A^t)_{ij} = A_{ji}$; $(aA + bB)^t = aA^t + bB^t$. A is **symmetric** if $A^t = A$ (so A is square). The symmetric matrices form a subspace of $\mathrm{M}_{n\times n}(\mathbb{F})$: $O^t = O$, $(A + B)^t = A + B$, $(aA)^t = aA$.

Example 1

$\mathsf{P}_n(\mathbb{F})$, the polynomials of degree $\leq n$, is a subspace of $\mathsf{P}(\mathbb{F})$.

Example 2

$C(\mathbb{R})$, the continuous real functions, is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Example 3

M is **diagonal** if $M_{ij} = 0$ for $i \neq j$. The diagonal $n \times n$ matrices form a subspace of $\mathrm{M}_{n\times n}(\mathbb{F})$.

Example 4

The **trace** is $\mathrm{tr}(M) = M_{11} + \cdots + M_{nn}$. It is linear, so the trace-zero matrices form a subspace.

Example 5

The matrices with nonnegative entries do not form a subspace of $\mathrm{M}_{m\times n}(\mathbb{R})$: multiplying by -1 leaves the set.

Theorem 1.4

Any intersection of subspaces of V is a subspace of V.

Proof. Let W be the intersection of a collection $\mathcal{C}$ of subspaces. Each contains 0, so $0 \in W$. If $x, y \in W$ and $a \in \mathbb{F}$, then $x + y$ and ax lie in every member of $\mathcal{C}$, hence in W. Apply Theorem 1.3. ■

A union of two subspaces is a subspace iff one contains the other. The **sum** $W_1 + W_2 = \{x + y : x \in W_1, y \in W_2\}$ is the smallest subspace containing both. V is the **direct sum** $W_1 \oplus W_2$ if $W_1 + W_2 = V$ and $W_1 \cap W_2 = \{0\}$.

1.4 Linear combinations and systems of linear equations

Motivation. Deciding whether a vector is a linear combination of others is solving a system of linear equations. The set of all linear combinations, the span, is the smallest subspace containing the given vectors.

Definition (Linear combination)

For nonempty $S \subseteq V$, v is a **linear combination** of vectors of S if $v = a_1u_1 + \cdots + a_nu_n$ for some finitely many $u_i \in S$ and scalars a_i (the **coefficients**). $0 = 0u$ is a linear combination of any nonempty S.

Example 1

Vitamin contents of foods are vectors in $\mathbb{R}^5$. A combination of foods has the corresponding linear combination of their vitamin vectors; e.g. the vitamin vector of some food is a sum of those of several others.

Solving systems by elimination. Three operations do not change the solution set of a system of linear equations: interchanging two equations, multiplying an equation by a nonzero constant, and adding a multiple of one equation to another. Using them, bring the system to a form where (i) the first nonzero coefficient of each equation is 1, (ii) that variable occurs in no other equation, and (iii) these leading variables move to the right from equation to equation. The solutions can then be read off; an equation $0 = c$ with $c \neq 0$ signals inconsistency.

Example 2

Is $2x^3 - 2x^2 + 12x - 6$ a linear combination of $x^3 - 2x^2 - 5x - 3$ and $3x^3 - 5x^2 - 4x - 9$? Comparing coefficients in $a(x^3 - 2x^2 - 5x - 3) + b(3x^3 - 5x^2 - 4x - 9)$ gives

$$a + 3b = 2, \quad -2a - 5b = -2, \quad -5a - 4b = 12, \quad -3a - 9b = -6,$$

with solution $a = -4$, $b = 2$. So yes. For $3x^3 - 2x^2 + 7x + 8$ the corresponding system reduces to an equation $0 = 1$, so it is not a linear combination.

Definition (Span)

The **span** $\operatorname{span}(S)$ of nonempty $S \subseteq V$ is the set of all linear combinations of vectors in S; $\operatorname{span}(\emptyset) = \{0\}$.

Theorem 1.5

$\operatorname{span}(S)$ is a subspace of V, and every subspace containing S contains $\operatorname{span}(S)$.

Proof. For $S = \emptyset$ this is clear. Otherwise $0 = 0z \in \operatorname{span}(S)$ for $z \in S$. If $x = \sum a_iu_i$ and $y = \sum b_jv_j$, then $x + y = \sum a_iu_i + \sum b_jv_j$ and $cx = \sum (ca_i)u_i$ are again linear combinations, so $\operatorname{span}(S)$ is a subspace by Theorem 1.3. A subspace $W \supseteq S$ is closed under addition and scalar multiplication, so it contains every $\sum a_iu_i$ with $u_i \in S$. ∎

Definition (Generating set)

S **generates** (**spans**) V if $\operatorname{span}(S) = V$.

Example 3

$(1,1,0),(1,0,1),(0,1,1)$ generate $\mathbb{R}^3$:

$$(a_1,a_2,a_3)=\tfrac{1}{2}(a_1+a_2-a_3)(1,1,0)+\tfrac{1}{2}(a_1-a_2+a_3)(1,0,1)+\tfrac{1}{2}(-a_1+a_2+a_3)(0,1,1).$$

Example 4

x^2+3x-2, $2x^2+5x-3$, $-x^2-4x+4$ generate $\mathsf{P}_2(\mathbb{R})$: every ax^2+bx+c is a combination of them (solve the 3×3 system).

Example 5

The matrices $\left(\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$ generate $\mathrm{M}_{2\times 2}(\mathbb{R})$. On the other hand, matrices all having equal diagonal entries cannot generate $\mathrm{M}_{2\times 2}(\mathbb{R})$, since every linear combination of them again has equal diagonal entries.

1.5 Linear dependence and linear independence

Motivation. A generating set may be wasteful: some vector may already be a combination of the others. Linear independence means no redundancy, i.e. 0 has only the trivial representation.

Definition (Linear dependence)

$S\subseteq V$ is **linearly dependent** if there are finitely many distinct $u_1,\dots,u_n\in S$ and scalars $a_1,\dots,a_n$, not all zero, with $a_1u_1+\cdots+a_nu_n=0$ (a **nontrivial representation** of 0). Any set containing 0 is dependent $(1\cdot 0=0)$.

Example 1

$S=\{(1,3,-4,2),(2,2,-4,0),(1,-3,2,-4),(-1,0,1,0)\}\subseteq\mathbb{R}^4$ is dependent: $4(1,3,-4,2)-3(2,2,-4,0)+2(1,-3,2,-4)+0(-1,0,1,0)=0$.

Example 2

In $\mathrm{M}_{2\times 3}(\mathbb{R})$, $\left(\begin{smallmatrix}1&-3&2\\-4&0&5\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}-3&7&4\\6&-2&-7\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}-2&3&11\\-1&-3&2\end{smallmatrix}\right)$ are dependent: $5M_1+3M_2-2M_3=O$.

Definition (Linear independence)

S is **linearly independent** if it is not linearly dependent: $\sum a_iu_i=0$ with distinct $u_i\in S$ forces all $a_i=0$.

$\emptyset$ is independent; $\{u\}$ is independent iff $u\neq 0$ (if $au=0$, $a\neq 0$, then $u=a^{-1}(au)=0$).

Example 3

$\{(1,0,0,-1),(0,1,0,-1),(0,0,1,-1),(0,0,0,1)\}$ is independent: $a_1(1,0,0,-1)+\cdots+a_4(0,0,0,1)=(a_1,a_2,a_3,a_4-a_1-a_2-a_3)=0$ forces $a_1=a_2=a_3=a_4=0$.

Example 4

For $p_k(x) = x^k + x^{k+1} + \cdots + x^n$ $(0 \le k \le n)$, $\{p_0, \ldots, p_n\}$ is independent: in $\sum a_k p_k = 0$ the coefficient of x^m is $a_0 + \cdots + a_m$, so $a_0 = 0$, then $a_1 = 0$, etc.

Theorem 1.6

If $S_1 \subseteq S_2 \subseteq V$ and S_1 is linearly dependent, so is S_2.

Proof. A nontrivial representation of 0 by vectors of S_1 is one by vectors of S_2. ■

Corollary

If $S_1 \subseteq S_2 \subseteq V$ and S_2 is linearly independent, so is S_1.

Proof. Contrapositive of Theorem 1.6. ■

Theorem 1.7

Let S be linearly independent and $v \notin S$. Then $S \cup \{v\}$ is linearly dependent iff $v \in \text{span}(S)$.

Proof.

Step 1 (only if). Take $a_1 u_1 + \cdots + a_n u_n = 0$ nontrivial with distinct $u_i \in S \cup \{v\}$. Since S is independent, some u_i, say u_1, equals v and $a_1 \neq 0$. Then $v = -a_1^{-1}(a_2 u_2 + \cdots + a_n u_n) \in \text{span}(S)$.

Step 2 (if). If $v = b_1 v_1 + \cdots + b_m v_m$ with distinct $v_i \in S$, then $b_1 v_1 + \cdots + b_m v_m + (-1)v = 0$ is nontrivial, with $v \neq v_i$. ■

1.6 Bases and dimension

Motivation. A basis is a generating set with no redundancy, and gives unique coordinates. The replacement theorem shows that independent sets never outnumber generating sets. So all bases have the same size, the dimension, and finite-dimensional spaces behave like $\mathbb{F}^n$.

Definition (Basis)

A **basis** β for V is a linearly independent subset that generates V.

Example 1

$\emptyset$ is a basis for the zero space.

Example 2

$e_1 = (1, 0, \ldots, 0), \ldots, e_n = (0, \ldots, 0, 1)$ form the **standard basis** of $\mathbb{F}^n$.

Example 3

The matrices E^{ij} (single 1 in position (i, j)) form a basis of $\text{M}_{m \times n}(\mathbb{F})$.

Example 4

$\{1, x, \dots, x^n\}$ is the **standard basis** of $\mathsf{P}_n(\mathbb{F})$.

Example 5

$\{1, x, x^2, \dots\}$ is a basis of $\mathsf{P}(\mathbb{F})$.

Theorem 1.8

$\beta = \{u_1, \dots, u_n\}$ is a basis for V iff each $v \in V$ can be written uniquely as $v = a_1u_1 + \cdots + a_nu_n$.

Proof.

Step 1 (only if). Existence holds since β generates V. If $v = \sum a_iu_i = \sum b_iu_i$, then $\sum(a_i - b_i)u_i = 0$, and independence gives $a_i = b_i$.

Step 2 (if) [Supplement]. β generates V. Uniqueness for $v = 0$ (with $0 = \sum 0u_i$) means $\sum a_iu_i = 0$ forces all $a_i = 0$, so β is independent. ■

So a basis of n vectors gives each v a unique n-tuple of coordinates; V "is" $\mathbb{F}^n$ (made precise in Section 2.4).

Theorem 1.9

If V is generated by a finite set S, then some subset of S is a basis for V. Hence V has a finite basis.

Proof. If $S = \emptyset$ or $S = \{0\}$, then $V = \{0\}$ and $\emptyset$ is a basis. Otherwise S contains some $u_1 \neq 0$, and $\{u_1\}$ is independent. Extend $\beta = \{u_1, \dots, u_k\} \subseteq S$ by vectors of S as long as it stays independent; since S is finite this stops at an independent $\beta \subseteq S$ such that $\beta \cup \{v\}$ is dependent for every $v \in S - \beta$. By Theorem 1.7, $S \subseteq \text{span}(\beta)$, so $V = \text{span}(S) \subseteq \text{span}(\beta)$ by Theorem 1.5. Hence β is a basis. ■

Example 6

From $S = \{(2,-3,5), (8,-12,20), (1,0,-2), (0,2,-1), (7,2,0)\}$ keep $(2,-3,5)$, discard $(8,-12,20) = 4(2,-3,5)$, keep $(1,0,-2)$ and $(0,2,-1)$ (each independent of the previous ones), and discard $(7,2,0) = 2(2,-3,5) + 3(1,0,-2) + 4(0,2,-1)$. So $\{(2,-3,5), (1,0,-2), (0,2,-1)\}$ is a basis of $\mathbb{R}^3$.

Theorem 1.10 (Replacement theorem)

Let V be generated by a set G of exactly n vectors and let L be linearly independent with exactly m vectors. Then $m \leq n$, and there is $H \subseteq G$ with exactly $n - m$ vectors such that $L \cup H$ generates V.

Proof. Induction on m; $m = 0$ is trivial with $H = G$. Let $L = \{v_1, \dots, v_{m+1}\}$ be independent. By Corollary , $\{v_1, \dots, v_m\}$ is independent, so by induction $m \leq n$ and some $\{u_1, \dots, u_{n-m}\} \subseteq G$ makes $\{v_1, \dots, v_m, u_1, \dots, u_{n-m}\}$ generate V. Write

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_{n-m}u_{n-m} = v_{m+1}. \tag{1.1}$$

If $n - m = 0$, or all $b_i = 0$, then $v_{m+1} \in \text{span}\{v_1, \dots, v_m\}$, contradicting Theorem 1.7. So $n \geq m + 1$ and some b_i, say b_1, is nonzero. Solving (1.1) for u_1 shows $u_1 \in \text{span}(L \cup H)$ with $H = \{u_2, \dots, u_{n-m}\}$

$(n-(m+1)$ vectors). Then $\operatorname{span}(L \cup H)$ contains $v_1, \ldots, v_m, u_1, \ldots, u_{n-m}$, hence equals V by Theorem 1.5. ■

Corollary 1

If V has a finite basis, every basis of V has the same number of vectors.

Proof. Let β be a basis with n vectors. Any independent γ has at most n vectors (apply Theorem 1.10 to finite subsets of γ), so γ is finite with $m \leq n$ vectors. Reversing roles, $n \leq m$. ■

Definition (Dimension)

V is **finite-dimensional** if it has a finite basis; the common number of vectors in a basis is its **dimension** $\dim(V)$. Otherwise V is **infinite-dimensional**.

Example 7

$\dim\{0\} = 0$.

Example 8

$\dim \mathbb{F}^n = n$.

Example 9

$\dim \mathrm{M}_{m\times n}(\mathbb{F}) = mn$.

Example 10

$\dim \mathsf{P}_n(\mathbb{F}) = n+1$.

Example 11

$\mathbb{C}$ over $\mathbb{C}$ has dimension 1 (basis $\{1\}$).

Example 12

$\mathbb{C}$ over $\mathbb{R}$ has dimension 2 (basis $\{1, i\}$). So the dimension depends on the field.

Corollary 2

Let $\dim(V) = n$.
(a) Every finite generating set has at least n vectors, and a generating set of exactly n vectors is a basis.
(b) Every linearly independent set of exactly n vectors is a basis.
(c) Every linearly independent subset of V extends to a basis.

Proof. Let β be a basis (n vectors).

(a) A finite generating set G contains a basis H by Theorem 1.9, and $|H| = n$ by Corollary 1. So $|G| \geq n$, and if $|G| = n$, then $G = H$.

(b) Apply Theorem 1.10 with $G = \beta$ and L independent, $|L| = n$: some $H \subseteq \beta$ with 0 vectors makes $L = L \cup H$ generate V.

(c) If L is independent with m vectors ($m \leq n$), Theorem 1.10 gives $H \subseteq \beta$ of $n - m$ vectors with $L \cup H$ generating V. Then $|L \cup H| \leq n$, and by (a) $L \cup H$ has exactly n vectors and is a basis. ■

Example 13

By Example 4 of Section 1.4 and (a), $\{x^2 + 3x - 2, 2x^2 + 5x - 3, -x^2 - 4x + 4\}$ is a basis of $\mathsf{P}_2(\mathbb{R})$.

Example 14

By Example 5 of Section 1.4 and (a), the four matrices there form a basis of $\mathrm{M}_{2\times 2}(\mathbb{R})$.

Example 15

By Example 3 of Section 1.5 and (b), $\{(1,0,0,-1), (0,1,0,-1), (0,0,1,-1), (0,0,0,1)\}$ is a basis of $\mathbb{R}^4$.

Example 16

By Example 4 of Section 1.5 and (b), $\{p_0, \ldots, p_n\}$ is a basis of $\mathsf{P}_n(\mathbb{F})$.

Summary. If $\dim V = n$: every basis has n vectors; independent sets have at most n vectors and extend to bases; generating sets have at least n vectors and shrink to bases; a set of exactly n vectors is a basis as soon as it is independent *or* generating.

Theorem 1.11

If W is a subspace of a finite-dimensional V, then W is finite-dimensional and $\dim W \leq \dim V$; if $\dim W = \dim V$, then $W = V$.

Proof. Let $\dim V = n$. If $W = \{0\}$, we are done. Otherwise choose $x_1 \neq 0$ in W and keep adding vectors $x_{k+1} \in W$ so that $\{x_1, \ldots, x_{k+1}\}$ stays independent. By Theorem 1.10 this stops after at most n steps, at an independent $\{x_1, \ldots, x_k\}$ to which no vector of W can be added. By Theorem 1.7 it generates W, so it is a basis, and $\dim W = k \leq n$. If $k = n$, it is a basis of V by Corollary 2(b), so $W = V$. ■

Example 17

$W = \{(a_1, \ldots, a_5) \in \mathbb{F}^5 : a_1 + a_3 + a_5 = 0, \ a_2 = a_4\}$ has basis $\{(-1,0,1,0,0), (-1,0,0,0,1), (0,1,0,1,0)\}$, so $\dim W = 3$.

Example 18

The diagonal $n \times n$ matrices have basis $\{E^{11}, \ldots, E^{nn}\}$; dimension n.

Example 19

The symmetric $n \times n$ matrices have basis $\{A^{ij} : 1 \le i \le j \le n\}$, where A^{ij} has 1 in positions (i,j) and (j,i) and 0 elsewhere; dimension $\frac{1}{2}n(n+1)$.

Corollary

If W is a subspace of a finite-dimensional V, any basis for W extends to a basis for V.

Proof. A basis of W is an independent subset of V; apply Corollary 2(c) of Theorem 1.10. ■

Example 20

The polynomials $a_{18}x^{18} + a_{16}x^{16} + \cdots + a_2x^2 + a_0$ form a subspace of $\mathsf{P}_{18}(\mathbb{F})$ with basis $\{1, x^2, \ldots, x^{18}\}$, dimension 10.

Subspaces of $\mathbb{R}^3$. Their dimensions are $0, 1, 2, 3$: the origin, lines through 0, planes through 0, and $\mathbb{R}^3$.

1.6.1 The Lagrange interpolation formula

Let $c_0, \ldots, c_n$ be distinct scalars of an infinite field $\mathbb{F}$, and

$$f_i(x) = \prod_{k \ne i} \frac{x - c_k}{c_i - c_k} \qquad (0 \le i \le n),$$

the **Lagrange polynomials**, of degree n, with $f_i(c_j) = \delta_{ij}$ (1 if $i = j$, else 0). They are independent: if $\sum a_i f_i = 0$, evaluating at c_j gives $a_j = 0$. Since $\dim \mathsf{P}_n(\mathbb{F}) = n+1$, $\beta = \{f_0, \ldots, f_n\}$ is a basis by Corollary 2(b). Evaluating $g = \sum b_i f_i$ at c_j gives $b_j = g(c_j)$, so

$$g = \sum_{i=0}^{n} g(c_i) f_i \qquad (g \in \mathsf{P}_n(\mathbb{F})),$$

the **Lagrange interpolation formula**. Hence for any $b_0, \ldots, b_n$ the unique polynomial of degree $\le n$ with $g(c_j) = b_j$ is $\sum b_i f_i$. In particular a nonzero polynomial of degree $\le n$ has at most n zeros.

1.7 Maximal linearly independent subsets *

Motivation. Infinite-dimensional spaces such as $\mathsf{P}(\mathbb{F})$ or $C(\mathbb{R})$ also have bases, but finding one in general needs an axiom of set theory. The Maximal Principle (equivalent to the Axiom of Choice) produces a maximal independent set, and maximality forces it to generate.

Definition (Maximal element; chain)

A member M of a family $\mathcal{F}$ of sets is **maximal** if no member of $\mathcal{F}$ other than M contains M. A collection $\mathcal{C}$ of sets is a **chain** if for all $A, B \in \mathcal{C}$, $A \subseteq B$ or $B \subseteq A$.

Example 1

In the power set of S, S is the (unique) maximal element.

Example 2

If S, T are disjoint and nonempty, the union of their power sets has two maximal elements S and T.

Example 3

The finite subsets of an infinite set S have no maximal element: add an element not yet present.

Example 4

$A_n = \{1, \ldots, n\}$, $n \geq 1$, form a chain.

Maximal Principle. Let $\mathcal{F}$ be a family of sets. If for every chain $\mathcal{C} \subseteq \mathcal{F}$ some member of $\mathcal{F}$ contains every member of $\mathcal{C}$, then $\mathcal{F}$ has a maximal member. (It is equivalent to the Axiom of Choice and is assumed here.)

Definition (Maximal linearly independent subset)

$B \subseteq S$ is a **maximal linearly independent subset of** S if (a) B is independent and (b) the only independent subset of S containing B is B.

Example 5

By Example 2 of Section 1.4, $\{x^3 - 2x^2 - 5x - 3, 3x^3 - 5x^2 - 4x - 9\}$ is a maximal linearly independent subset of $S = \{2x^3 - 2x^2 + 12x - 6, x^3 - 2x^2 - 5x - 3, 3x^3 - 5x^2 - 4x - 9\}$; so is any two-element subset of S.

A basis β of V is a maximal independent subset of V: for $v \notin \beta$, $\beta \cup \{v\}$ is dependent by Theorem 1.7.

Theorem 1.12

If S generates V and β is a maximal linearly independent subset of S, then β is a basis for V.

Proof. It suffices to show $S \subseteq \text{span}(\beta)$ (Theorem 1.5). If $v \in S - \text{span}(\beta)$, then $\beta \cup \{v\}$ is independent by Theorem 1.7, contradicting maximality. ■

Theorem 1.13

If S is a linearly independent subset of V, some maximal linearly independent subset of V contains S.

Proof. Let $\mathcal{F}$ be the family of independent subsets of V containing S. For a chain $\mathcal{C} \subseteq \mathcal{F}$, let U be the union of its members (or $U = S$ if $\mathcal{C} = \emptyset$). Then $U \supseteq S$ and U contains every member of $\mathcal{C}$. U is independent: if $u_1, \ldots, u_n \in U$ are distinct with $\sum a_i u_i = 0$, say $u_i \in A_i \in \mathcal{C}$, then since $\mathcal{C}$ is a chain, one A_k contains all A_i. Since A_k is independent, all $a_i = 0$. So $U \in \mathcal{F}$, and the Maximal Principle gives a maximal member of $\mathcal{F}$. It is a maximal independent subset of V, since any independent set containing it also contains S. ■

Corollary

Every vector space has a basis.

Proof. Apply Theorem 1.13 to $S = \emptyset$, and Theorem 1.12 with $S = V$. ■

All bases of an infinite-dimensional space also have the same cardinality (cited).